

Projection-and-Pooling for Non-Convex Isotonic Regression on Finite Partial Orders*

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Abstract

We study isotonic regression on a finite partially ordered set in which fitted values satisfy monotonicity constraints. In convex-loss settings, this leads to a standard isotonic projection problem with a pooling interpretation. We extend this framework to a class of non-convex objectives. For separable criteria with a shared nonlinear component, we show that the resulting non-convex order-restricted problem admits an exact two-step representation: first compute a weighted isotonic projection under the same partial order, then perform a scalar maximization within each level set of the projected solution. The projection determines the pooling structure, while the nonlinear component determines the fitted values on each pooled block. A counterexample shows that this decomposition fails without the shared-component structure. We establish fixed-grid consistency under a stability condition ensuring local robustness of the population pooling partition. The framework covers penalized least squares, Bregman divergences, exponential-family likelihoods, and scalar non-convex perturbations such as quartic potentials. Simulations and an intraday Invesco QQQ ETF upper-tail quantile application illustrate the method in tail-risk settings with sharp regime changes. Quartic projection-and-pooling estimators improve recovery of monotone quantile surfaces, with the largest gains concentrated in stress regions and extreme upper-tail regimes, while preserving the imposed partial-order constraints.

Keywords: isotonic regression; partial orders; non-convex optimization; shape-constrained estimation; fixed-grid quantile function

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1 Introduction

Shape restrictions arise naturally in many statistical applications. In a dose–response study, the mean response or event probability is often expected to increase with dose. In financial risk measurement, an upper-tail loss quantile should not decrease when both recent volatility and recent drawdown increase. Monotonicity constraints of this kind are commonly handled through isotonic regression and related order-restricted methods (Ayer et al.; 1955; Barlow et al.; 1972). This paper studies isotonic estimation on a finite partially ordered set. Each index in the set corresponds to one fitted value, and the fitted values are required to respect a monotonicity relation induced by the partial order. In many applications, these indices correspond to cells of a discretized covariate grid; throughout the paper, we therefore occasionally refer to them as *grid cells*. The focus is on finite-grid problems, where the order structure is fixed and known.

In standard least-squares isotonic regression, the estimation problem is convex and has a well-known projection and pooling structure: the isotonic projection groups observations into *level sets*, meaning maximal subsets of indices that share a common fitted value. Many modern shape-constrained methods retain this convex structure, including constrained least squares, isotonic distributional regression, non-crossing quantile regression, and related approaches (Beran and Dümbgen; 2010; Turlach; 2005; Pya and Wood; 2015; Henzi et al.; 2021; Dai et al.; 2023; Shim et al.; 2009). In some applications, however, the objective function naturally contains nonlinear components that make the problem non-convex. Examples include penalized criteria, likelihood-based formulations under monotone parametrizations, or controlled nonlinear perturbations designed to reduce oversmoothing in regions with abrupt changes. A natural question is therefore the following:

Can a non-convex order-restricted problem still retain the computational and geometric structure of ordinary isotonic regression?

The main contribution of this paper is to show that, for a broad class of non-convex objectives, the answer is yes. Specifically, we show that the global solution can still be obtained by first computing an ordinary weighted isotonic projection on the original partial order and then solving separate one-dimensional optimization problems within the pooled level sets determined by that projection. Thus, despite the non-convexity of the objective, the only step requiring joint optimization across indices remains the familiar convex isotonic projection. After the pooling structure is determined, the remaining optimization decomposes into separate one-dimensional problems within each pooled level set.

More precisely, we consider a class of order-constrained optimization problems over a finite index set $I = \{1, \dots, n\}$ equipped with a partial order $\preceq$ (i.e., a reflexive, transitive, and antisymmetric relation), with one fitted value x_i attached to each index $i \in I$. Let $E \subset I \times I$ denote a collection of ordered pairs generating the partial order $\preceq$. The monotonicity constraints require

$$x_i \leq x_j \quad \text{whenever } i \preceq j.$$

We henceforth work directly with the partial order $\preceq$ and suppress the generating set E .

Let $x = (x_1, \dots, x_n) \in X^n$, where $X \subset \mathbb{R}$ is an interval or a finite set, and consider problems of the form

$$\max_{x \in X^n} \sum_{i \in I} f_i(x_i) \quad \text{subject to} \quad x_i \leq x_j \text{ whenever } i \preceq j. \quad (1.1)$$

The objective is separable across indices, with $f_i : X \rightarrow \mathbb{R}$ acting on the scalar coordinate x_i , while dependence across coordinates is induced solely through the order constraints.

Our focus is on a structural class of coordinate-wise objective functions of the form

$$f_i(x_i) = k_i a(x_i) + m_i b(x_i) + \ell_i, \quad m_i > 0, \quad (1.2)$$

where a is a strictly increasing scalar function, b is a scalar nonlinear function common to all indices, and k_i , m_i , and ℓ_i are index-specific coefficients. We refer to b as the *common nonlinear component*, as it contributes the same nonlinear transformation of x_i in every coordinate objective. This shared nonlinear structure is the key restriction enabling the reduction scheme developed in this paper. In particular, b governs the curvature of the objective, including possible non-concavity, while a determines the scale on which the monotonicity constraint acts. A simple motivating example is bounded isotonic regression with a common quartic perturbation, where each coordinate objective consists of an index-specific linear term combined with a shared nonlinear component. For sufficiently large perturbation strength, the resulting scalar objective becomes non-concave, illustrating that the model class extends beyond standard convex isotonic regression settings. Appendix G discusses the mechanism-design motivation for the separable objective class (1.2) under order constraints, through ironing and its connection to isotonic regression in discrete settings.

A key feature of the formulation in (1.1)–(1.2) is that all dependence across coordinates arises solely through the monotonicity constraints. The objective is otherwise separable across indices, with each coordinate sharing the same nonlinear component $b(\cdot)$. This separation suggests that the effect of the order constraints can be isolated from the coordinate-wise nonlinear structure. In particular, it can be seen that a weighted isotonic projection of the rescaled coefficients induces a partition of the index set into groups. Conditional on this partition, the problem reduces to independent scalar optimization problems within each group.

The structural class in (1.2) extends beyond squared-error criteria. This includes, in particular, penalized least squares problems with common penalties applied to the fitted values, as well as a range of likelihood-based and divergence-based criteria that admit monotone reparametrizations, including certain exponential-family models and Bregman-type losses.

Our work is related to two main literatures. The first concerns shape-constrained statistical estimation. Existing methods for isotonic regression, monotone smoothing, additive models, isotonic distributional regression, and constrained quantile estimation are typically based on convex formulations (Beran and Dümbgen; 2010; Turlach; 2005; Pya and Wood; 2015; Henzi et al.; 2021; Dai et al.; 2023; Shim et al.; 2009). Computational work in this literature includes design-based approaches to isotonic regression and non-crossing quantile regression (Colubi et al.; 2018; Shim et al.; 2009). A parallel line of work studies risk, adaptation, and asymptotic properties of isotonic estimators as the order structure grows (Han et al.; 2019; Deng and Zhang; 2020). In contrast, we keep the index set

fixed and study how a non-convex criterion can nevertheless inherit the geometry of standard isotonic regression through a projection-based reduction.

The second related literature concerns non-convex isotonic regression and isotone optimization. Existing approaches often rely on special order structures, dynamic programming, pruning methods, or convex reformulations (Lim; 2018; Bach; 2018; Wang et al.; 2022). Our result differs in that, for the structural class in (1.2), all interaction across coordinates enters through a standard isotonic projection step. After this step, the problem reduces to independent one-dimensional optimization problems over the induced level sets. The proof combines a monotonicity-based single-crossing argument with geometric properties of isotonic projections on partial orders.

Against this background, the paper makes three contributions.

1. We prove that a projection-based reduction yields the global solution of (1.1) for the structural class (1.2) on arbitrary finite partial orders.
2. We show that the shared nonlinear component is essential: the reduction can fail when nonlinear terms vary freely across indices.
3. We establish fixed-grid statistical properties of the resulting estimator, including consistency under stable pooling structures.

The numerical sections illustrate the proposed projection-based mechanism through quantile estimation problems. The simulation study considers monotone fixed-grid quantile functions that exhibit regions of rapid change over short segments of the grid, corresponding to localized steepness in the underlying signal. We compare the standard isotonic projection estimator as a baseline with an estimator that applies nonlinear adjustments after the projection step to capture remaining nonlinear variation within pooled groups. Next, the empirical application studies upper-tail quantile estimation for intraday Invesco QQQ ETF returns across ordered volatility and drawdown grid cells and across multiple tail probability levels. Particular attention is given to the *stress region*, corresponding to the upper-right portion of the grid where both recent volatility and drawdown are elevated.

The remainder of the paper is organized as follows. Section 2 presents the general non-convex isotonic formulation and the corresponding projection-based characterization. Section 3 discusses computation and examples of statistical criteria covered by our theorem. Section 4 reports the simulation study. Section 5 presents the empirical QQQ ETF application. Section 6 concludes.

2 Projection and pooling

This section presents the main projection-based characterization result. We show that, under the structural assumption introduced above, the solution of the non-convex isotonic problem can be characterized through a weighted isotonic projection that summarizes the effect of the order constraints.

We consider the optimization problem (1.1)–(1.2) with the following assumption.

Assumption 2.1 (Regularity of a , b , and X). The set X is a nonempty compact interval or a finite grid, a is continuous and strictly increasing, b is upper semicontinuous, and $m_i > 0$ for all i .

Remark 2.1. The common scale a is primarily a normalization. Because a is strictly increasing, the change of variables $z_i = a(x_i)$ preserves all order constraints and transforms the objective into $k_i z_i + m_i \tilde{b}(z_i) + \ell_i$, where $\tilde{b} = b \circ a^{-1}$. Thus, up to an order-preserving reparametrization, it suffices to consider the case $a(x_i) = x_i$. By contrast, the shared nonlinear component condition is substantive: it is the shape restriction needed for the single-crossing comparison below. It requires all indices to have the same curvature, including any non-concavity, while allowing the linear coefficients and weights to vary across indices.

2.1 Characterization

We begin by isolating the part of the objective that interacts with the order constraints. The function a is a strictly increasing transformation of the decision variable and preserves the ordering of values within each coordinate. As a result, it does not affect how the ordering constraints shape the optimization problem. In contrast, heterogeneity across indices enters through the coefficients k_i and the weights m_i , which determine how each coordinate contributes to the objective. The nonlinear component b governs the curvature of each coordinate-wise objective but remains common across all indices. To separate these roles, define the rescaled coefficients

$$g_i = \frac{k_i}{m_i}.$$

This normalization isolates the index-specific component of the objective from the scaling induced by m_i . Since m_i determines both the contribution of each coordinate in the objective and the weighting structure used in the projection step below, the relevant ordering information is captured by g_i .

The ordering structure is then represented through a weighted isotonic projection of g onto the partial order $\preceq$, defined as

$$g^* := \arg \min_{y \in \mathbb{R}^n} \sum_{i \in I} m_i (y_i - g_i)^2 \quad \text{subject to} \quad y_i \leq y_j \text{ whenever } i \preceq j. \quad (2.1)$$

Let g^* denote the projection. This solution induces a partition of the index set into maximal subsets on which g^* is constant. We refer to these subsets as the *level sets*, and to the resulting partition as the *pooling structure*.

The projection can also be characterized through standard first-order optimality conditions. To express this, define the order cone

$$\mathcal{K} = \{y \in \mathbb{R}^n : y_i \leq y_j \text{ whenever } i \preceq j\}. \quad (2.2)$$

With respect to the weighted inner product $\langle y, z \rangle_m = \sum_{i \in I} m_i y_i z_i$, the projection satisfies the orthogonality condition

$$\langle g - g^*, z \rangle_m \leq 0 \quad \text{for all } z \in \mathcal{K}. \quad (2.3)$$

Moreover, the residual $g - g^*$ is orthogonal (in the weighted sense) to any vector that is constant on the level sets of g^* . Equivalently, for any function ψ defined on the finite set of distinct values taken

by g^* ,

$$\sum_{i \in I} (g_i - g_i^*) \psi(g_i^*) m_i = 0. \quad (2.4)$$

This identity follows from the first-order optimality condition of the weighted isotonic projection. Indeed, consider perturbations that add a common shift to all coordinates within any given level set of g^* while preserving the relative ordering between distinct level sets. For sufficiently small perturbations, the resulting vectors continue to satisfy the order constraints, and applying the first-order optimality condition in these directions yields (2.4).

We now use the projection g^* to construct an auxiliary separable optimization problem that preserves the structure of the original problem while replacing the index-specific coefficients by order-compatible ratios. Define modified coefficients by

$$\bar{k}_i := m_i g_i^*,$$

and define modified coordinate functions

$$\bar{f}_i : X \rightarrow \mathbb{R}, \quad \bar{f}_i(t) := \bar{k}_i a(t) + m_i b(t) + \ell_i.$$

The corresponding separable objective is

$$\bar{f}(x) := \sum_{i \in I} \bar{f}_i(x_i).$$

The modified objective $\bar{f}$ retains the same separable structure as the original objective f , while its coefficient ratios satisfy $\bar{k}_i/m_i = g_i^*$. By construction of the isotonic projection, these ratios are consistent with the partial order.

We therefore consider the coordinatewise maximizer sets

$$\Phi_i := \arg \max_{t \in X} \bar{f}_i(t), \quad \Phi := \{x \in X^n : x_i \in \Phi_i \text{ for all } i \in I\}. \quad (2.5)$$

Under Assumption 2.1, each Φ_i is nonempty and compact by the Weierstrass extreme value theorem. The remaining question is whether selections from these coordinatewise maximizer sets can be combined into a vector satisfying the original order constraints.

The next lemma shows that larger coefficient ratios produce ordered maximizers. Applied to the modified coordinate functions $\bar{f}_i$, this transfers the ordering of g^* to the selected values from Φ_i .

Lemma 2.1. *If*

$$x_i \in \arg \max_{t \in X} \bar{f}_i(t) \quad \text{and} \quad x_j \in \arg \max_{t \in X} \bar{f}_j(t),$$

then

$$g_i < g_j \quad \Rightarrow \quad x_i \leq x_j.$$

If $g_i = g_j$, then the two maximizer sets are identical.

Proof. Since x_i maximizes f_i over X ,

$$k_i a(x_i) + m_i b(x_i) + \ell_i \geq k_i a(x_j) + m_i b(x_j) + \ell_i.$$

Similarly, since x_j maximizes f_j over X ,

$$k_j a(x_j) + m_j b(x_j) + \ell_j \geq k_j a(x_i) + m_j b(x_i) + \ell_j.$$

Dividing the first inequality by m_i and the second by m_j , and then adding, yields

$$(g_j - g_i)\{a(x_j) - a(x_i)\} \geq 0.$$

If $g_i < g_j$, then $a(x_j) \geq a(x_i)$. Since a is strictly increasing, it follows that $x_i \leq x_j$. If $g_i = g_j$, then the normalized objectives f_i/m_i and f_j/m_j differ only by an additive constant. Hence they have the same maximizer set over X . $\square$

We now use the projection identity to relate the original objective f to the modified objective $\bar{f}$. The role of the projected vector g^* is to replace the original coefficient ratios g_i by order-compatible values while preserving enough structure to characterize the maximizers of the original problem.

Recall the order cone in (2.2). If $x \in X^n$ satisfies the order constraints, then $(a(x_i))_{i \in I} \in \mathcal{K}$, because a is strictly increasing. Therefore, using the projection relation (2.3),

$$f(x) - \bar{f}(x) = \sum_{i \in I} (g_i - g_i^*) a(x_i) m_i = \langle g - g^*, (a(x_i))_{i \in I} \rangle_m \leq 0.$$

Thus the modified objective $\bar{f}$ dominates the original objective f over the order-constrained set. Moreover, the two objectives agree at a given order-constrained vector x if and only if

$$\sum_{i \in I} (g_i - g_i^*) a(x_i) m_i = 0. \tag{PG}$$

We refer to condition (PG) as the *pooling property*.

The following theorem shows that the comparison between f and $\bar{f}$ is exact. Solving the original problem therefore reduces to selecting coordinatewise maximizers of the modified objective in a way that makes the upper bound above attain equality.

Theorem 2.1 (Projection and pooling characterization). *Let $\mathcal{M} := \{x \in X^n : x_i \leq x_j, i \preceq j\}$ be the feasible set of the original isotonic problem. Under Assumption 2.1, an order-constrained vector $x \in \mathcal{M}$ solves the original problem (1.1)–(1.2) if and only if $x \in \Phi$, where Φ is defined in (2.5) and x satisfies the pooling property (PG).*

The proof is given in Appendix A. The theorem has an immediate constructive implication. Although the equality condition (PG) is global, it is automatically satisfied whenever the selected values

are constant on the level sets of the projection g^* . Specifically, a sufficient condition is

$$g_i^* = g_j^* \quad \Rightarrow \quad x_i = x_j, \quad (\text{PG2})$$

because under (PG2), the quantity $a(x_i)$ depends only on the common projected value g_i^* . Hence one may write $a(x_i) = \psi(g_i^*)$ for some function ψ on the set of distinct projected values, and (2.4) then implies the pooling property (PG).

It remains to verify the order constraints. Lemma 2.1, applied to the modified coordinate functions $\bar{f}_i$, shows that selections from the sets Φ_i inherit the ordering of the projected values g_i^* . Consequently, any selection $x \in \Phi$ that is constant on the level sets of g^* satisfies the order constraints and also satisfies (PG). By Theorem 2.1, every such selection solves the original problem (1.1)–(1.2).

In particular, the componentwise minimal and maximal selections provide two canonical solutions. Define

$$x_{*i} := \min \Phi_i, \quad x_i^* := \max \Phi_i, \quad i \in I.$$

Then

$$x_* := (x_{*i})_{i \in I}, \quad x^* := (x_i^*)_{i \in I},$$

are both order-constrained and satisfy (PG2). Hence both solve the original problem.

Corollary 2.1. *The vectors x_* and x^* both solve the problem (1.1)–(1.2).*

2.2 Shared nonlinear component

The shared nonlinear component assumption is essential to the reduction developed above. It ensures that the coefficient ratio $g_i = k_i/m_i$ determines the ordering of coordinatewise maximizers across indices. If the nonlinear component is allowed to vary arbitrarily across indices, Lemma 2.1 may fail.

Example 2.1 (Failure with index-specific nonlinear components). Let $I = \{1, 2\}$ with $1 \preceq 2$, let $X = \{0, 1\}$, and let $a(t) = t$ for $t \in X$. Suppose

$$m_1 = m_2 = 1, \quad k_1 = 0, \quad k_2 = 1, \quad \ell_1 = \ell_2 = 0,$$

so that $g_1 < g_2$. Define index-specific nonlinear functions by

$$b_1(0) = 0, \quad b_1(1) = 2, \quad b_2(0) = 0, \quad b_2(1) = -2.$$

Then the first coordinate objective is maximized at $t = 1$, whereas the second is maximized at $t = 0$. Thus, despite $g_1 < g_2$, the corresponding maximizers are ordered in the reverse direction. Consequently, the projection-based reduction cannot depend only on the ratios g_i when the nonlinear components vary freely across indices.

Some index-specific nonlinearities can nevertheless be absorbed into the structural class. Suppose

$$b_i(t) = \rho_i b(t) + \delta_i a(t) + \gamma_i, \quad \rho_i > 0.$$

Then $k_i a(t) + m_i b(t) + \ell_i$ can be rewritten as $k'_i a(t) + m'_i b(t) + \ell'_i$, where

$$k'_i := k_i + m_i \delta_i, \quad m'_i := m_i \rho_i, \quad \ell'_i := \ell_i + m_i \gamma_i.$$

Hence the problem remains within the structural class after reparametrization.

Beyond such transformations, exact characterization would generally require additional compatibility conditions linking the projection g^* to the coordinatewise maximizer sets Φ_i . Consequently, objectives involving residual trimming, capped losses, graph smoothing penalties, or spline roughness penalties typically fall outside the scope of Theorem 2.1 unless analyzed separately.

2.3 Fixed-grid statistical consistency and rates

The projection-and-pooling characterization is purely deterministic. In statistical applications, we consider a fixed finite partially ordered index set I and a corresponding collection of population-level parameters $(g_i^0, m_i^0)_{i \in I}$. These are deterministic quantities indexed by I that define the target fixed-grid function and are not functions of the observed sample once the data-generating process is fixed. Given data, we construct empirical counterparts $(\hat{g}_i, \hat{m}_i)_{i \in I}$ and apply the same projection-and-pooling operator to obtain an estimator on the same index set I . The analysis therefore concerns how perturbations in these finite-dimensional estimates propagate through the deterministic mapping, with the population fixed-grid function induced by $(g_i^0, m_i^0)_{i \in I}$ serving as the target object. No asymptotic regime is considered in which the index set expands or the grid approximates a continuum.

Let $g^0 = (g_i^0)_{i \in I}$ and $m^0 = (m_i^0)_{i \in I}$ denote the population index-level ratios and weights, with $m_i^0 > 0$. The weights are defined up to a common scaling factor; in empirical cell-based settings, the normalized empirical weights $\hat{m}_i$ may be taken as cell frequencies n_i/N .

For a shared nonlinear component b , define the scalar maximizer correspondence

$$\mathcal{T}(u) := \arg \max_{t \in X} \{u a(t) + b(t)\}.$$

When this set is a singleton, we write $T(u)$ for its unique element.

The next assumption specifies the regularity conditions required to transfer estimation error through the deterministic projection-and-pooling operator. These include uniform consistency of the primitive estimators, bounded and strictly positive weights, and stability of the scalar maximization map.

Assumption 2.2 (Fixed-grid empirical estimates and stable scalar maximizers). Let the finite partially ordered index set $(I, \preceq)$ be fixed. The population weights satisfy $0 < \underline{m} \leq m_i^0 \leq \bar{m} < \infty$ for all $i \in I$. For a deterministic sequence $r_N \downarrow 0$, the empirical estimators satisfy

$$\max_{i \in I} |\hat{g}_i - g_i^0| = O_p(r_N), \quad \max_{i \in I} |\hat{m}_i - m_i^0| = O_p(r_N).$$

Recall the order cone $\mathcal{K} = \{y \in \mathbb{R}^n : y_i \leq y_j \text{ whenever } i \preceq j\}$. Define the weighted quadratic criteria

$$\hat{Q}(y) := \sum_{i \in I} \hat{m}_i (y_i - \hat{g}_i)^2, \quad Q_0(y) := \sum_{i \in I} m_i^0 (y_i - g_i^0)^2,$$

and the corresponding isotonic projections

$$\hat{g}^* := \arg \min_{y \in \mathcal{K}} \hat{Q}(y), \quad g^{0,*} := \arg \min_{y \in \mathcal{K}} Q_0(y).$$

Finally, assume that for every u in a neighborhood of $\{g_i^{0,*} : i \in I\}$, the maximizer set $\mathcal{T}(u)$ is a singleton and the map T is locally Lipschitz on that neighborhood.

Theorem 2.2 (Fixed-grid consistency and rate). *Under Assumption 2.2, the isotonic projections satisfy $\max_{i \in I} |\hat{g}_i^* - g_i^{0,*}| = O_p(r_N)$. Moreover, if $\hat{x}_i := T(\hat{g}_i^*)$ and $x_i^0 := T(g_i^{0,*})$, then $\max_{i \in I} |\hat{x}_i - x_i^0| = O_p(r_N)$. Thus, the projection-and-pooling estimator inherits the uniform rate of the primitive index-level estimators on the fixed finite partially ordered set.*

We note that the weights $(\hat{m}_i)_{i \in I}$ enter through the definition of the weighted isotonic projection and affect the projection operator, although the resulting rate is stated in terms of the induced projection $\hat{g}^*$.

Corollary 2.2 (Cell quantile rate). *Let Z denote the underlying covariate (state) random variable taking values in a space partitioned into fixed cells $\{\mathcal{C}_i\}_{i \in I}$. Each index $i \in I$ corresponds to a cell $\mathcal{C}_i$, and let $\hat{g}_i$ denote the empirical quantile computed within that cell.*

If the cell probabilities $\pi_i := \mathbb{P}(Z \in \mathcal{C}_i)$ are uniformly bounded away from zero and the conditional distribution of the outcome Y given $Z \in \mathcal{C}_i$ admits a density that is positive and continuous at the target quantile, then under independent sampling,

$$\max_{i \in I} |\hat{g}_i - g_i^0| = O_p(N^{-1/2}), \quad \max_{i \in I} |\hat{m}_i - m_i^0| = O_p(N^{-1/2}).$$

The same rate holds under stationarity and suitable mixing conditions that ensure a joint empirical quantile central limit theorem. Consequently, if T is locally Lipschitz on a neighborhood of $\{g_i^{0,} : i \in I\}$, then*

$$\max_{i \in I} |\hat{x}_i - x_i^0| = O_p(N^{-1/2}).$$

The proof of Theorem 2.2 is given in Appendix B. The argument proceeds by expressing both the empirical and population projections through their variational inequalities on the same fixed order cone. A quadratic perturbation bound for the weighted least-squares criterion controls $\hat{g}^* - g^{0,*}$ in terms of $(\hat{g}_i - g_i^0)$ and $(\hat{m}_i - m_i^0)$, and the local Lipschitz property of T transfers this stability to the final projection-and-pooling estimator. Corollary 2.2 then follows by combining standard fixed-cell empirical quantile rates with Theorem 2.2.

These results establish fixed-grid convergence rates but do not describe the finer stochastic behavior of the isotonic projection beyond consistency. A more detailed limit theory would characterize the exact random fluctuations of the projected vector after rescaling, but this level of precision is not needed for our purposes. Instead, we focus on a regular regime in which the population pooling structure is stable in the sense that small perturbations in the input should not change which indices are grouped together by the projection. This stability is guaranteed when different pooled blocks are sufficiently separated in the population solution. The following condition formalizes this idea in a finite partial-order setting and plays the role of the usual strict block separation condition in one-dimensional isotonic regression.

Assumption 2.3 (Strict pooling margins). Let $\mathcal{B}^* = \{B_1, \dots, B_R\}$ be the pooling partition induced by $g^{0,*}$, and write $g_i^{0,*} = c_r$ for $i \in B_r$. We impose two separation conditions ensuring that this partition is stable under small perturbations.

- (i) **Separation between distinct pooled blocks.** All distinct ordered level sets are strictly separated:

$$\eta_{\text{sep}} = \min\{c_s - c_r : B_r \prec B_s, r \neq s\} > 0,$$

with the minimum over an empty set interpreted as $+\infty$. Here $B_r \prec B_s$ means that some element of B_r is below some element of B_s in the partial order.

- (ii) **Internal stability of each pooled block.** Within each level set B_r , no proper lower subset has vanishing weighted imbalance:

$$\eta_{\text{pool}} = \min_r \min_{L \in \mathcal{L}_r} \sum_{i \in L} m_i^0 (g_i^0 - c_r) > 0,$$

where $\mathcal{L}_r$ denotes the collection of nonempty proper lower sets of the partial order restricted to B_r .

Proposition 2.1 (Pooling partition stability). *Suppose Assumptions 2.2 and 2.3 hold, and that*

$$\max_{i \in I} |\hat{g}_i - g_i^0| + \max_{i \in I} |\hat{m}_i - m_i^0| \xrightarrow{P} 0 \quad \text{as } N \rightarrow \infty,$$

where N denotes the sample size used to construct the empirical quantities $(\hat{g}_i, \hat{m}_i)_{i \in I}$ while the index set I and the underlying partition remain fixed. If $\hat{\mathcal{B}}$ is the pooling partition induced by $\hat{g}^*$, then $\mathbb{P}(\hat{\mathcal{B}} = \mathcal{B}^*) \rightarrow 1$ as $N \rightarrow \infty$.

Remark 2.2 (Implications of stable pooling). Proposition 2.1 implies that, in the regular fixed-grid regime considered here, the empirical projection eventually reproduces the same pooling partition as the population projection with high probability. Once this occurs, the projection-and-pooling estimator behaves locally like a finite-dimensional averaging procedure within the stable pooled blocks. As a result, the first-order stochastic behavior of the estimator is inherited from the underlying empirical cellwise estimators.

3 Computation and examples of statistical criteria

The projection-and-pooling characterization in Theorem 2.1 leads directly to a practical computational procedure. The only step involving the partial order is the weighted isotonic projection (2.1). Once this projection is computed, the remaining optimization reduces to a collection of one-dimensional scalar maximization problems, one for each level set of the projected vector g^* . Algorithm 1 summarizes the procedure before we discuss examples of statistical criteria covered by the theorem.

Algorithm 1 Projection and pooling algorithm

1. Compute the ratios $g_i = k_i/m_i$ and solve the weighted isotonic projection problem (2.1) for g^* .
 2. Form the modified coefficients and component objectives $\bar{k}_i = m_i g_i^*$, $\bar{f}_i(t) = \bar{k}_i a(t) + m_i b(t) + \ell_i$.
 3. Identify the level sets $L_r = \{i \in I : g_i^* = c_r\}$ of g^* . For each level set, compute the maximizer set of $c_r a(t) + b(t)$ over $t \in X$.
 4. Assign a common representative maximizer to all indices in the same level set. The component-wise minimal and maximal selections yield canonical optimizers.
-

The first step of Algorithm 1 can be carried out using any reliable weighted isotonic regression routine for the given partial order. For a chain order, the projection problem (2.1) is solved in $O(n)$ time by the pool adjacent violators algorithm (PAVA) (Ayer et al.; 1955). In this case, the solution also admits the cumulative sum diagram characterization of Barlow et al. (1972): if

$$W_i = \sum_{j \leq i} m_j, \quad G_i = \sum_{j \leq i} m_j g_j,$$

then the projected values g_i^* are given by the slopes of the greatest convex minorant of the cumulative sum diagram (W_i, G_i) .

For more general order structures, the appropriate projection routine depends on the geometry of the partial order. Rooted tree orders can be handled by minimum violator methods (Thompson; 1962; Pardalos and Xue; 1999), while two-dimensional grid orders admit dynamic programming approaches (Spouge et al.; 2003). General finite partial orders can be solved through max flow/min cut formulations (Maxwell and Muckstadt; 1985; Hochbaum and Queyranne; 2003), yielding complexity of order $O(n|E| \log(n^2/|E|))$. A broader survey of efficient algorithms for L_p isotonic regression is given by Stout (2019).

Once the isotonic projection has been computed, the remaining steps of Algorithm 1 are purely one-dimensional. Depending on the form of the shared nonlinear component b , the scalar maximization problems may admit closed-form solutions or can be solved by grid search or standard scalar numerical optimization routines. This separation between the projection step and the scalar maximization step is computationally useful: the same order-restricted projection can be reused across different specifications of the shared nonlinear component b .

We next discuss statistical criteria covered by Theorem 2.1. Since the theorem is formulated in maximization form, a loss-minimization problem can be rewritten by negating the loss and omitting terms that do not depend on the fitted value. After this transformation, the criterion must take the structural form as in (1.2). Penalized least squares provides a leading example. Let $\lambda \geq 0$ denote a tuning parameter controlling the strength of the penalty, and let $P : X \rightarrow \mathbb{R}$ be a scalar penalty function applied coordinatewise to the fitted values. Consider

$$L_\lambda(g, t) = \frac{1}{2}(t - g)^2 + \lambda P(t), \quad t \in X.$$

Negating the loss gives $-L_\lambda(g, t) = gt - t^2/2 - \lambda P(t) - g^2/2$. Thus the criterion fits the structural form with

$$a(t) = t, \quad k_i = m_i g_i, \quad b(t) = -\frac{1}{2}t^2 - \lambda P(t).$$

In the simulation and empirical sections, we use a quartic potential as a simple controlled way to introduce nonconcavity while preserving the shared nonlinear component structure:

$$\sum_i m_i \left\{ g_i t_i - \frac{1}{2} t_i^2 + \lambda t_i^4 \right\}, \quad t_i \in [0, 1]. \quad (3.1)$$

Here $b_\lambda(t) = -t^2/2 + \lambda t^4$, which becomes nonconcave on $[0, 1]$ once $\lambda > 1/12$. The quartic term can amplify larger fitted values but may also push solutions toward the boundary of the parameter space. The empirical implementation therefore uses a validation rule that excludes boundary solutions.

For this specification, the scalar post-projection step reduces to maximizing $ut - t^2/2 + \lambda t^4$ over $t \in [0, 1]$. The fixed-grid consistency result applies for any fixed λ such that the relevant scalar maximizer is locally unique and locally Lipschitz in u . The validation rule is intended to avoid boundary regimes where this local continuity may become fragile in finite samples.

We also note that the same decomposition also clarifies which procedures fall outside the scope of Theorem 2.1. Hard trimming and capped residual losses are natural robust alternatives, with related computational work on trimmed robust regression (Pitsoulis and Zioutas; 2010). However, these methods modify or selectively remove observation-level contributions rather than applying a common scalar penalty to each fitted coordinate. Consequently, they lead to related order-constrained nonconvex optimization problems, but they do not arise directly from the projection-and-pooling characterization developed here.

However, the current framework covers several other important classes of statistical criteria, including Bregman divergence losses and exponential-family likelihoods based on index-level sufficient statistics. For a differentiable strictly convex function $\phi : \mathcal{X} \rightarrow \mathbb{R}$ defined on an interval $\mathcal{X} \subset \mathbb{R}$ with $\phi'(\mathcal{X}) \subset \mathbb{R}$, the associated Bregman divergence is

$$-D_\phi(g, t) = g\phi'(t) + \phi(t) - t\phi'(t) - \phi(g), \quad g, t \in \mathcal{X}.$$

Hence the structural representation holds with

$$a(t) = \phi'(t), \quad b(t) = \phi(t) - t\phi'(t),$$

so that $a : \mathcal{X} \rightarrow \mathbb{R}$ and $b : \mathcal{X} \rightarrow \mathbb{R}$ (Bregman; 1967; Banerjee et al.; 2005).

Similarly, a canonical exponential-family log-likelihood contribution of the form

$$m_i \{ g_i \eta(t_i) - A(\eta(t_i)) \}, \quad t_i \in \mathcal{T},$$

with $\mathcal{T} \subset \mathbb{R}$ and canonical parameter map $\eta : \mathcal{T} \rightarrow \mathbb{R}$, fits the framework with

$$a(t) = \eta(t), \quad b(t) = -A(\eta(t)), \quad t \in \mathcal{T}.$$

Here A is the log-partition function, and g_i denotes the sufficient statistic at index i normalized by m_i . Appendix E presents a binary-response likelihood example on the logit scale illustrating this exponential-family formulation.

4 Simulation study

We use Monte Carlo simulations to study the projection-and-pooling estimator in a fixed-grid quantile estimation setting. The design is motivated by upper-tail loss estimation in which risk increases with observable state variables, while the underlying quantile surface may exhibit sharp transitions across regions of the grid. The objective belongs to the structural class covered by Theorem 2.1, so the simulation directly examines the projection-and-pooling mechanism developed in the theory.

The target consists of two upper-tail quantile surfaces on a fixed grid, corresponding to the 95% and 99% quantiles of the loss. The goal is to compare the standard convex isotonic projection with quartic post-projection transformations in a setting where convex smoothing may attenuate sharp changes across regions of the state space. We consider both a fixed mild quartic perturbation and a quartic specification selected by validation, paralleling the tuning procedure used in the empirical illustration below. All reported results are averaged over $B = 100$ Monte Carlo replications.

4.1 Design

We discretize two state variables into an ordered $K \times L$ grid with $K = L = 12$. Each grid cell is indexed by $(k, m) \in \{1, \dots, K\} \times \{1, \dots, L\}$, and represents a pair of discretized state levels, for which we consider discretized volatility level and discretized drawdown level, respectively. Here, drawdown refers to the distance from the historical maximum to the current level, which is commonly used as a measure of recent downside movement. These variables are interpreted as observable conditions describing the environment under which losses are evaluated, rather than as structural economic objects.

The fitted quantity is a positive Value-at-Risk (VaR) magnitude, equivalently an upper-tail quantile of the loss distribution. Let $(\alpha_1, \alpha_2) = (0.95, 0.99)$ denote the target quantile levels. For each grid cell and each level α_t ($t = 1, 2$), we target the corresponding population upper-tail quantile

$$x_{k,m,t}^* \in X = [0, \bar{x}],$$

which represents the true (unobserved) VaR level, where larger values correspond to larger downside risk. The grid is equipped with the product partial order $(k, m) \preceq (k', m')$ if and only if $k \leq k'$ and $m \leq m'$. This imposes the monotonicity constraints

$$x_{k,m,t} \leq x_{k+1,m,t}, \quad x_{k,m,t} \leq x_{k,m+1,t}.$$

We additionally impose

$$x_{k,m,1} \leq x_{k,m,2},$$

so that the more extreme upper-tail quantile level is constrained to have at least as large a VaR

magnitude. Together, these inequalities define a finite partial order on the index set (k, m, t) .

The true function is monotone in both grid indices but exhibits sharp discrete shifts across certain regions. Let $(k_0, m_0) = (8, 8)$ denote threshold locations that separate low- and high-stress regions of the grid, where “stress” refers to the level of the underlying state variables (here, discretized volatility and drawdown) associated with more adverse loss outcomes. For each $t \in \{1, 2\}$, define

$$x_{k,m,t}^* = 0.35 \frac{k}{K} + 0.25 \frac{m}{L} + \Delta_{\sigma,t} \mathbf{1}\{k \geq k_0\} + \Delta_{\ell,t} \mathbf{1}\{m \geq m_0\} + \Delta_{\sigma\ell,t} \mathbf{1}\{k \geq k_0, m \geq m_0\},$$

where the indicator terms introduce level shifts once either or both thresholds are crossed and these shifts are larger for the more extreme tail, with

$$(\Delta_{\sigma,1}, \Delta_{\ell,1}, \Delta_{\sigma\ell,1}) = (0.50, 0.35, 0.40), \quad (\Delta_{\sigma,2}, \Delta_{\ell,2}, \Delta_{\sigma\ell,2}) = (0.90, 0.60, 0.70),$$

so that $x_{k,m,1}^* \leq x_{k,m,2}^*$ holds for all (k, m) .

The data-generating mechanism is specified at the grid-cell level by a location-scale model for returns. For each (k, m) , we generate

$$R_{k,m} = \mu_{k,m} + \sigma_{k,m} Z, \quad Z \sim t_5,$$

where $\mu_{k,m} \in \mathbb{R}$ and $\sigma_{k,m} > 0$ are chosen so that the induced upper-tail loss quantiles match the target surface $x_{k,m,t}^*$ for $t \in \{1, 2\}$. For each grid cell, we generate $N_{\text{train}} = 60$ training observations and compute empirical loss quantile estimates $\hat{x}_{k,m,t}$. We enforce the within-cell ordering across quantile levels by replacing

$$\hat{x}_{k,m,2} \leftarrow \max\{\hat{x}_{k,m,2}, \hat{x}_{k,m,1}\}.$$

To mimic finite-sample distortion near high-risk regions, we introduce rare negative jumps only in the training sample, with probabilities that increase near the threshold boundaries and in the region where both grid indices exceed their cutoff levels. An independent validation sample of size $N_{\text{val}} = 120$ per grid cell is generated without these perturbations and is used solely to select the tuning parameter λ . A clean test sample of size $N_{\text{test}} = 200$ per grid cell is used for final evaluation. Throughout, the population target remains x^* , so mean squared error measures recovery error for the quantile function rather than changes in the underlying target.

The convex and quartic estimators differ only through the shared nonlinear component b . Let $i \equiv (k, m, t)$ index grid cells and quantile levels, and set $g_i = \hat{x}_{k,m,t}$, $a(x) = x$. The convex specification corresponds to $b_0(x) = -x^2/2$.

For numerical stability and consistency with the empirical section later, the projection and scalar maximization are implemented on the normalized scale $z = x/\bar{x}_{\text{sim}} \in [0, 1]$, where $\bar{x}_{\text{sim}} = 1.05 \max_{k,m,t} x_{k,m,t}^*$. On this scale, the quartic specification takes the form $uz - z^2/2 + \lambda z^4$, $z \in [0, 1]$, where u denotes the projection output rescaled to the unit interval, $u = x^*/\bar{x}_{\text{sim}}$, with x^* the (pre-normalization) projected value. Here, we set $\lambda = 0.05$ as a mild perturbation of the convex benchmark, yielding a smooth and numerically stable scalar maximization problem on $[0, 1]$.

Method	Grid MSE	Stress region MSE	JumpErr _{0.95}	JumpErr _{0.99}	QL _{0.95}	QL _{0.99}
Convex ($\lambda = 0$)	0.2898 (0.0140)	1.0654 (0.0707)	0.2486	0.6290	0.0957	0.0381
Quartic, fixed $\lambda = 0.05$	0.2680 (0.0148)	0.9489 (0.0760)	0.2396	0.6045	0.0949	0.0368
Quartic, validation choice	0.2189 (0.0172)	0.6778 (0.0927)	0.2242	0.5489	0.0935	0.0343

Table 1: Simulation results for the synthetic fixed-grid quantile surface experiment. Standard deviations are shown in parentheses for the mean squared error criteria.

By contrast, the validation-based specification selects λ from the grid

$$\Lambda = \{0, 0.005, 0.01, 0.02, 0.05, 0.075, 0.10, 0.125, 0.15\}$$

using an independent validation sample. The selection rule favors the smallest value achieving near-minimal validation error in the high-risk region while excluding specifications that produce boundary solutions.

For each replication, we first solve the convex isotonic projection (2.1) under the finite partial order to obtain g^* . We then compute

$$\hat{x}_i^\lambda = T_\lambda(g_i^*),$$

where

$$\mathcal{T}_\lambda(u) := \operatorname{argmax}_{z \in [0,1]} \left\{ uz - \frac{1}{2}z^2 + \lambda z^4 \right\}, \quad T_\lambda(u) := \max \mathcal{T}_\lambda(u).$$

Thus, $T_\lambda(u)$ denotes the maximal scalar maximizer associated with the projected value u . The fitted values are then transformed back to the original scale for evaluation.

The first evaluation criterion is the global mean squared error (MSE) of the estimated fixed-grid quantile surface:

$$\text{MSE} = \frac{1}{2KL} \sum_{t=1}^2 \sum_{k=1}^K \sum_{m=1}^L (\hat{x}_{k,m,t} - x_{k,m,t}^*)^2.$$

The second criterion is the same MSE restricted to the (high-stress) region $\{(k, m) : k \geq k_0, m \geq m_0\}$, where both stress indicators exceed their threshold levels. The third criterion measures recovery of the regime-shift magnitudes at the threshold boundaries, denoted by $\text{JumpErr}_{\alpha_t}$. Specifically, it evaluates how accurately the estimator reproduces the imposed jump components $\Delta_{\sigma,t}, \Delta_{\ell,t}, \Delta_{\sigma\ell,t}$, which govern the changes in the quantile surface across the stress boundaries. Smaller values of $\text{JumpErr}_{\alpha_t}$ therefore indicate better recovery of the sharp structural transitions built into the data-generating mechanism. The final criterion is the out-of-sample quantile loss (QL_{α_t}) evaluated on the clean test sample.

4.2 Results

Table 1 reports Monte Carlo averages for the main simulation comparison; standard deviations are shown in parentheses for the two grid mean squared error criteria. Relative to the convex benchmark, the fixed quartic perturbation improves performance across all evaluation criteria. The validation-based quartic specification produces larger gains, with the strongest improvements appearing in the more extreme 0.99 tail.

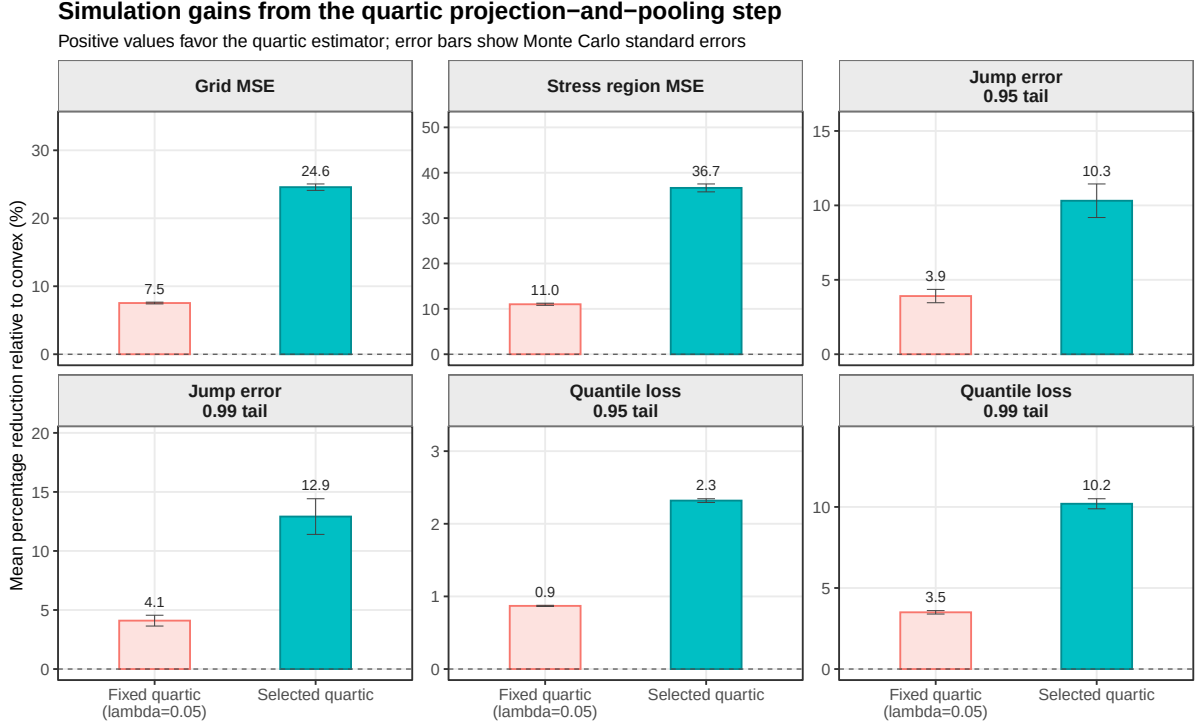


Figure 1: Simulation gains relative to the convex isotonic projection.

The largest improvements occur in the stress-region MSE criteria, indicating that the quartic post-projection transformation more accurately recovers the sharp increases in the target quantile surface near the stress thresholds. The reductions in $\text{JumpErr}_{0.95}$ and $\text{JumpErr}_{0.99}$ further show that the quartic specifications better capture the magnitudes of the regime shifts built into the data-generating mechanism, whereas the convex projection tends to smooth these transitions. The gains are most pronounced for the 0.99 tail, where the target surface contains larger threshold-induced shifts and stronger nonlinear stress effects.

Figure 1 visualizes the same performance improvements. Both quartic specifications outperform the convex projection across all six evaluation criteria, with the validation-based specification producing the largest gains, particularly for global grid MSE, stress-region MSE, and the 0.99 tail criteria. These results suggest that a moderate non-concave post-projection transformation can improve recovery of fixed-grid upper-tail quantile surfaces with sharp regime changes, while the fixed $\lambda = 0.05$ specification mainly serves as a stable reference perturbation.

The validation procedure selects a positive value of λ in nearly all replications, implying that the preferred specification is typically non-concave near the upper boundary of the normalized domain. All fitted surfaces satisfy the order constraints up to numerical tolerance. Appendix D repeats the simulation over grid sizes $G \in \{8, 12, 16\}$ and threshold locations $k_0 = m_0 = \lceil \rho G \rceil$ with $\rho \in \{0.60, 0.67, 0.75\}$. The validation-based quartic estimator continues to reduce both global and stress-region MSE across all nine designs, with the largest gains again occurring in the 0.99 tail.

5 Invesco QQQ ETF upper-tail quantile surface illustration

This section presents an empirical illustration of fixed-grid upper-tail quantile estimation under order restrictions. The goal is not a full dynamic VaR backtesting analysis, but rather a comparison of order-restricted estimators on discretized volatility–drawdown regimes. Specifically, after grouping observations into ordered grid cells determined by recent volatility and drawdown conditions, we ask whether the quartic post-projection transformation improves out-of-sample upper-tail quantile estimation relative to the convex isotonic projection.

5.1 Data and estimators

The empirical illustration uses five-minute prices for the Invesco QQQ ETF, tracking the NASDAQ-100 index, from January 2, 2020 through December 31, 2025, restricted to regular trading hours. For each horizon $h \in \{30, 60, 120\}$ minutes, the forward loss is defined by $L_{t,h} = -\log(P_{t+h}/P_t)$, where P_t denotes the observed price at time t . Forward losses are computed only from within regular trading hours in order to avoid overnight effects.

The ordered covariates are based on recent intraday market conditions. The first covariate,

$$X_{1,t} = \left(\sum_{j=1}^{12} r_{t-j}^2 \right)^{1/2},$$

is the lagged one-hour realized volatility, where $r_t = \log(P_t/P_{t-1})$ denotes the five-minute return. Larger values of $X_{1,t}$ correspond to periods of elevated short-run price variability. The second covariate,

$$X_{2,t} = \max_{0 \leq u \leq v \leq 12} \{\log P_{t-v} - \log P_{t-u}\},$$

is the lagged one-hour realized drawdown, measuring the largest cumulative price decline over the recent one-hour window. Larger drawdowns correspond to more severe recent downward price movements.

The index set of grid cells is equipped with a partial order induced by the two covariates $(X_{1,t}, X_{2,t})$, where $(x_1, x_2) \preceq (x'_1, x'_2)$ if and only if $x_1 \leq x'_1$ and $x_2 \leq x'_2$. Under this partial order, a grid cell with both higher recent volatility and a larger recent drawdown is considered higher in the order and therefore should not be assigned a smaller forward upper-tail loss quantile. This imposes a monotonicity constraint consistent with the idea that more severe recent market conditions correspond to stochastically larger tail risk.

The primary horizon is $h = 60$ minutes, while the 30- and 120-minute horizons are included as sensitivity checks. Table 2 reports the effective sample sizes together with empirical upper-tail loss quantiles, showing that the scale of tail losses increases with the forecast horizon.

For each rolling training window, each ordered covariate is discretized into bins, and the Cartesian product of these bins forms a $G \times G$ grid with $G \in \{8, 10\}$. Each grid element is a grid cell. For cell c and quantile level $\alpha \in \{0.95, 0.975, 0.99\}$, the empirical estimate is denoted by $\hat{x}_{c,\alpha}$. Monotonicity over the ordered grid and non-crossing across quantile levels are imposed as order constraints on the combined grid–quantile index set.

Horizon	Observations	$Q_{0.95}$	$Q_{0.975}$	$Q_{0.99}$
30 min	91,964	0.00448	0.00614	0.00858
60 min	82,904	0.00633	0.00869	0.01209
120 min	64,791	0.00905	0.01231	0.01643

Table 2: Summary statistics for the QQQ intraday sample. The reported quantiles are empirical upper-tail quantiles of the forward loss $L_{t,h}$.

The convex estimator is the weighted isotonic projection. The quartic estimator applies Algorithm 1 after scaling each training window by its 0.999 empirical loss quantile and using $b_\lambda(x) = -x^2/2 + \lambda x^4$ on $[0, 1]$. A positive λ is allowed only if it passes a validation rule designed to exclude boundary solutions, as imposed in the previous simulation study section.

Performance is evaluated on two-month rolling test windows. The primary criterion is grid MSE for the upper-tail quantile function,

$$\text{GridMSE} = \frac{\sum_{c,\alpha} m_c^{\text{test}} (\hat{x}_{c,\alpha} - \hat{x}_{c,\alpha}^{\text{test}})^2}{\sum_c m_c^{\text{test}}},$$

where m_c^{test} denotes the number of test observations in cell c , and $\hat{x}_{c,\alpha}^{\text{test}}$ is the corresponding empirical test-window quantile. Stress region MSE restricts the sum to the upper-right portion of the ordered grid. Observation-level performance is additionally evaluated using the standard check (pinball) loss, aggregated over test observations; the lower value indicates better calibration of the upper-tail probability. Finally, we report paired fold-by-fold comparisons across rolling test windows, where performance differences are computed within each out-of-sample window and then aggregated across windows, to assess whether improvements are persistent over time rather than driven by a small number of specific episodes.

5.2 Results

Table 3 reports grid-level performance across 22 rolling test windows. Entries report percentage reductions relative to the convex isotonic projection, with positive values indicating improvements. Across all horizons and grid resolutions, the quartic estimator consistently improves upon the convex benchmark.

For the primary 60-minute horizon on the 8×8 grid, global grid MSE decreases by 5.32%, while stress-region MSE decreases by 7.49%. Improvements are largest at the 30-minute horizon and remain stable at 120 minutes, indicating that gains are not driven by a specific forecast horizon. The selected quartic coefficient parameter λ is moderate on average.

Horizon	Grid	Grid MSE reduction	Stress region MSE reduction	Mean λ
30	8	6.23%	11.53%	0.094
30	10	5.62%	8.84%	0.084
60	8	5.32%	7.49%	0.082
60	10	5.43%	7.63%	0.076
120	8	4.13%	5.37%	0.080
120	10	3.77%	4.87%	0.078

Table 3: QQQ grid MSE reductions for the fixed-grid upper-tail quantile function.

Observation-level pinball (i.e., check-function) losses exhibit a similar pattern. Table 4 reports results for the 8×8 grid, with positive entries indicating reductions relative to the convex isotonic projection. Improvements are modest at the lower tail level $\alpha = 0.95$, but become progressively larger as the quantile level increases, with the strongest gains in the extreme tail and in the stress region.

For the primary one-hour horizon at $\alpha = 0.99$, the overall pinball loss decreases by 5.54%, while the stress-region pinball loss decreases by 8.97%, indicating improved accuracy specifically in high-risk regimes.

Horizon	Quantile	Pinball reduction	Stress region pinball reduction
30	0.950	1.04%	1.95%
30	0.975	2.48%	4.62%
30	0.990	6.47%	12.04%
60	0.950	0.79%	1.35%
60	0.975	1.98%	3.42%
60	0.990	5.54%	8.97%
120	0.950	0.98%	1.21%
120	0.975	2.34%	3.13%
120	0.990	6.00%	8.36%

Table 4: Observation-level pinball loss reductions for the 8×8 grid.

Table 5 reports fold-level paired evidence based on one-sided Wilcoxon signed-rank tests computed over the 22 rolling test windows. For each fold, the paired difference is defined as the convex loss minus the quartic loss, so positive values indicate improvement of the quartic estimator. The one-sided test evaluates whether these differences are systematically positive across folds. The last column reports the fold-by-fold outcome counts of the paired comparison, where each fold is classified as “improvement” if the quartic loss is smaller, “equal” if the losses coincide up to numerical precision, and “worse” otherwise. These tests are reported as rolling-window diagnostic summaries rather than as a formal substitute for a full dynamic VaR backtesting framework.

Horizon	Grid	Grid MSE	Stress region MSE	Stress region pinball _{0.99}	Imp./eq./worse
30	8	0.0004	0.0025	0.0012	15/7/0
30	10	0.0013	0.0085	0.0013	13/8/1
60	8	0.0032	0.0252	0.0072	11/9/2
60	10	0.0040	0.0151	0.0032	11/9/2
120	8	0.0127	0.0458	0.0043	8/10/4
120	10	0.0127	0.0458	0.0043	8/10/4

Table 5: Fold-level one-sided Wilcoxon signed-rank test statistics based on paired loss differences across rolling windows. Identical values in the 120-minute rows reflect discreteness of the signed-rank statistic, even though the underlying fold-level difference sequences differ.

Figure 2 summarizes the corresponding patterns for global grid MSE, stress region MSE, and stress-region pinball loss at the $\alpha = 0.99$ quantile. For $G = 8$, a positive λ is selected in 15, 13, and 12 of the 22 folds at the 30-, 60-, and 120-minute horizons, respectively, while the remaining folds revert to the convex projection. This indicates that the improvements arise from an adaptive post-projection correction applied in a subset of regimes, rather than from uniformly imposing non-convexity across all windows. The gains are most pronounced in stress-region cells and at more extreme tail levels, consistent with the simulation design.

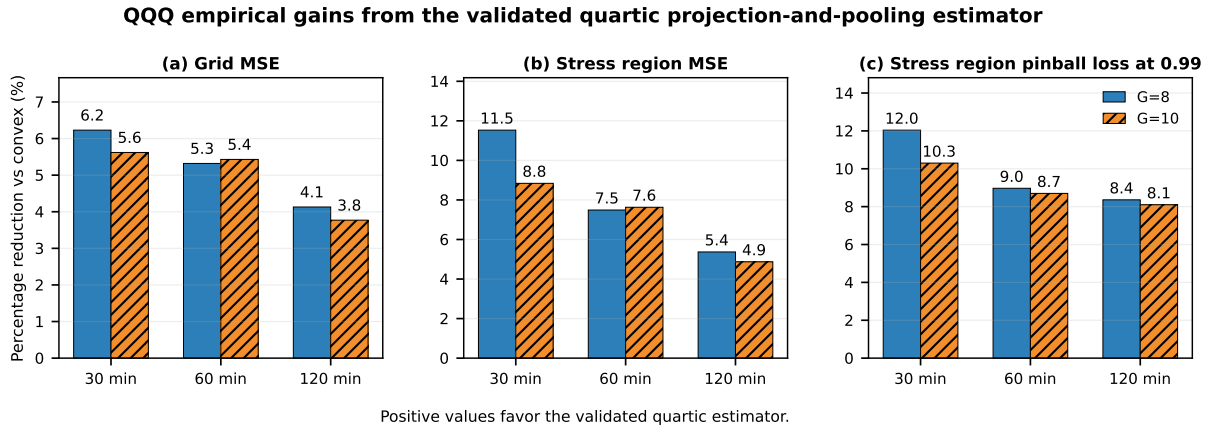


Figure 2: Fold-level gains of the quartic estimator relative to the convex isotonic projection across rolling windows.

6 Discussion

This paper proposes a practical method for non-convex isotonic regression on finite partial orders that preserves the computational structure of standard isotonic regression. When the objective shares a common nonlinear component b and an increasing component a , the estimator is obtained in two steps: a weighted isotonic projection under the same order constraints, followed by scalar maximization within each pooled block. As a result, the order-restricted step remains a standard isotonic projection.

The key condition is the shared nonlinear component across indices. This ensures that ordering is governed by the ratio structure k_i/m_i , so pooling remains valid after projection. When this structure

is violated, as shown in Section 2.2, the decomposition fails. This also clarifies why common practical adjustments such as trimming or winsorization do not fall directly under this framework.

The empirical results indicate usefulness in tail-risk settings. In simulations, the quartic projection-and-pooling estimator better captures sharp changes in the quantile surface, especially in stress regions and the upper tail. In the QQQ ETF application, the validation-selected correction consistently improves the convex isotonic benchmark across horizons, with gains concentrated in high-volatility and high-drawdown regimes. Overall, allowing a mild shared non-convex adjustment after isotonic projection improves fit when the underlying risk surface has sharp regime shifts, provided tuning avoids boundary instability.

Several extensions are natural in applied work. These include finer grids for large-scale risk systems, likelihood-based models for discrete outcomes, and robustness adjustments such as censoring. These directions would require extending the projection-based structure to handle cell-specific deviations that are not consistent across the grid, and thus cannot be represented through the shared nonlinear component assumed in the current framework.

Data Availability Statement

The code for the simulation study and empirical analysis is available at <https://github.com/crbaek/nonconvex-isotonic>. The empirical analysis uses proprietary Refinitiv data, which cannot be redistributed by the authors. Researchers with access to Refinitiv can obtain the data directly from the provider and reproduce the empirical results using the published code.

A Proof of Theorem 2.1

We prove both directions. For any $x \in \mathcal{M}$,

$$f(x) - \bar{f}(x) = \sum_{i \in I} (k_i - \bar{k}_i) a(x_i) = \sum_{i \in I} (g_i - g_i^*) a(x_i) m_i.$$

Since $x \in \mathcal{M}$ and a is strictly increasing, the vector $(a(x_1), \dots, a(x_n))$ belongs to $\mathcal{K}$. Hence, by (2.3),

$$\sum_{i \in I} (g_i - g_i^*) a(x_i) m_i \leq 0.$$

Therefore $f(x) \leq \bar{f}(x)$ for all $x \in \mathcal{M}$, implying

$$\sup_{x \in \mathcal{M}} f(x) \leq \sup_{x \in \mathcal{M}} \bar{f}(x) \leq \sup_{x \in X^n} \bar{f}(x).$$

For sufficiency, suppose $x \in \mathcal{M}$, $x \in \Phi$, and x satisfies (PG). Since $x \in \Phi$, it maximizes $\bar{f}$ over X^n , and therefore also over $\mathcal{M}$. Moreover, (PG) implies $f(x) = \bar{f}(x)$. Thus the upper bound above is attained at x , so x solves (1.1)–(1.2).

For necessity, suppose $x \in \mathcal{M}$ solves (1.1)–(1.2). Define $x_i^\dagger = \max \Phi_i$ for each $i \in I$. By Lemma 2.1, applied to the modified functions $\bar{f}_i$, we have $x^\dagger \in \mathcal{M}$: if $g_i^* < g_j^*$ then $x_i^\dagger \leq x_j^\dagger$, while if $g_i^* = g_j^*$ then

$\Phi_i = \Phi_j$ and therefore $x_i^\dagger = x_j^\dagger$.

The equality of maximizer sets within each level set of g^* implies (PG2), and hence (PG) by (2.4). Since $x^\dagger \in \Phi$, the sufficiency argument above shows that $x^\dagger$ solves (1.1)–(1.2) and maximizes $\bar{f}$ over X^n . Consequently,

$$\sup_{z \in \mathcal{M}} f(z) = f(x^\dagger) = \bar{f}(x^\dagger) = \sup_{z \in X^n} \bar{f}(z).$$

For the original optimizer x ,

$$f(x) \leq \bar{f}(x) \leq \sup_{z \in X^n} \bar{f}(z) = \sup_{z \in \mathcal{M}} f(z) = f(x).$$

Hence all inequalities are equalities. Therefore $\bar{f}(x) = \sup_{z \in X^n} \bar{f}(z)$, which implies $x \in \Phi$, and $f(x) - \bar{f}(x) = 0$, which is exactly the pooling property (PG). This completes the proof.

B Proof of Theorem 2.2

Write $y^0 = g^{0,*}$ and $\hat{y} = \hat{g}^*$. Let $\hat{a} = \min_i \hat{g}_i$ and $\hat{b} = \max_i \hat{g}_i$. If a feasible vector $y \in \mathcal{K}$ is truncated componentwise to the interval $[\hat{a}, \hat{b}]$, the order constraints remain satisfied and the criterion $\sum_i \hat{m}_i (y_i - \hat{g}_i)^2$ cannot increase. Hence $\hat{y}$ is bounded in probability. The same argument applies to the population projection y^0 . The population projection satisfies

$$\sum_{i \in I} m_i^0 (y_i^0 - g_i^0) (y_i - y_i^0) \geq 0, \quad y \in \mathcal{K},$$

which implies the quadratic growth bound

$$Q_0(y) - Q_0(y^0) \geq \underline{m} \|y - y^0\|_2^2, \quad y \in \mathcal{K}, \quad (\text{B.1})$$

where $Q_0(y) = \sum_i m_i^0 (y_i - g_i^0)^2$.

On any fixed compact set containing $\hat{y}$ and y^0 with probability tending to one,

$$\left| \{\widehat{Q}(y) - Q_0(y)\} - \{\widehat{Q}(y^0) - Q_0(y^0)\} \right| \leq O_p(r_N) \|y - y^0\|_2, \quad (\text{B.2})$$

uniformly in y , where $\widehat{Q}(y) = \sum_i \hat{m}_i (y_i - \hat{g}_i)^2$. By optimality of the empirical projection,

$$\widehat{Q}(\hat{y}) \leq \widehat{Q}(y^0). \quad (\text{B.3})$$

Combining (B.1), (B.2), and (B.3) gives $\underline{m} \|\hat{y} - y^0\|_2^2 \leq O_p(r_N) \|\hat{y} - y^0\|_2$. Therefore $\|\hat{y} - y^0\|_2 = O_p(r_N)$. Since the index set I is finite, this implies $\max_i |\hat{g}_i^* - g_i^{0,*}| = O_p(r_N)$. The local Lipschitz property of T at the finitely many projected values then yields $\max_i |T(\hat{g}_i^*) - T(g_i^{0,*})| = O_p(r_N)$.

C Proof of Proposition 2.1

Proof. We use the finite-dimensional characterization of weighted isotonic projections through their level sets. For a fixed pooling partition, the projection is feasible when the corresponding level-set values satisfy the quotient partial order. Optimality within a level set is characterized by nonnegativity of the directional derivative for every feasible downward split along a lower set of the induced order. Equivalently, this is the KKT, or min-max, characterization of weighted isotonic regression restricted to a constant level set.

For a population level set B_r with projected value c_r , the relevant inequalities are

$$\sum_{i \in L} m_i^0 (g_i^0 - c_r) \geq 0$$

for every lower set L of the order induced on B_r , with equality when $L = B_r$. Under Assumption 2.3, every nonempty proper lower set satisfies the inequality with a strict margin. Hence the strict residual margin $\eta_{\text{pool}} > 0$ places the population level set in the interior of the region where the same pooling inequalities remain valid.

Similarly, the separation margin $\eta_{\text{sep}} > 0$ ensures that projected values assigned to distinct ordered level sets cannot cross under sufficiently small perturbations. Since there are only finitely many level sets and lower sets, both strict margins remain positive whenever

$$\max_i |\hat{g}_i - g_i^0| + \max_i |\hat{m}_i - m_i^0| < \varepsilon$$

for sufficiently small deterministic $\varepsilon > 0$.

On this event, the original level-set partition continues to satisfy the optimality inequalities for the empirical weighted projection, and the empirical level-set values preserve the quotient order. Hence the empirical projection has the same pooling partition as the population projection. Since the perturbation bound holds with probability tending to one, $P(\hat{\mathcal{B}} = \mathcal{B}^*) \rightarrow 1$. $\square$

D Simulation robustness across grids and locations of sharp changes

Table 6 reports robustness checks discussed in Section 4. We vary the grid size $G \in \{8, 12, 16\}$ and the location of the sharp changes, $k_0 = m_0 = \lceil \rho G \rceil$, with $\rho \in \{0.60, 0.67, 0.75\}$. Each design is based on $B = 100$ Monte Carlo replications and uses the same validation rule as in the main simulation.

The reported entries are mean percentage reductions relative to the convex isotonic projection. The gains remain positive across all nine designs, indicating that the main results are not driven by the baseline 12×12 grid or by the specific location of the regime shifts (e.g., $(8, 8)$ in the baseline design).

G	ρ	Grid MSE	Stress region MSE	JumpErr _{0.99}	QL _{0.99}
8	0.60	27.24%	37.29%	12.90%	11.89%
12	0.60	24.31%	36.06%	10.31%	10.17%
16	0.60	25.02%	35.93%	10.41%	11.65%
8	0.67	23.49%	37.61%	13.08%	7.76%
12	0.67	21.24%	36.07%	11.74%	7.57%
16	0.67	22.95%	36.17%	11.73%	9.62%
8	0.75	21.94%	35.13%	11.68%	7.81%
12	0.75	20.75%	35.25%	8.56%	7.59%
16	0.75	20.52%	36.07%	10.16%	7.70%

Table 6: Robustness of simulation results for the quartic estimator selected by the validation rule. Entries report mean percentage reductions relative to the convex isotonic projection.

E Binary response likelihood example

This appendix provides an additional likelihood-based illustration of Theorem 2.1 in a fixed-grid binary-response setting. The design is intentionally stylized and is included to verify that the projection-and-pooling mechanism extends beyond least-squares-type objectives. All results are based on $B = 200$ Monte Carlo replications.

Let s_i denote the number of successes in cell i out of n_i Bernoulli trials, and let $x_i = \log\{p_i/(1-p_i)\}$ be the logit parameter. The quartic binomial criterion is defined on a compact interval $\mathcal{X} = [\eta_L, \eta_U] \subset \mathbb{R}$ as

$$f_i(t) = s_i t - n_i \log(1 + e^t) + n_i \lambda R(t), \quad t \in \mathcal{X},$$

where $R(t) = \{(t - \eta_L)/(\eta_U - \eta_L)\}^4$. This fits the separable structure with

$$a(t) = t, \quad b_\lambda(t) = -\log(1 + e^t) + \lambda R(t), \quad k_i = s_i, \quad m_i = n_i, \quad t \in \mathcal{X}.$$

Table 7 shows that the quartic specification yields substantially lower errors than the convex binomial isotonic projection. The largest gains occur in the stress region, consistent with the presence of a sharp change in latent logit risk, where the convex projection tends to over-smooth. The quartic term reduces this smoothing by allowing stronger separation across regimes. This example should therefore be interpreted as a controlled likelihood illustration of the framework rather than a forecasting benchmark for binary outcomes. Across replications, the selected λ is either 1 or 1.5, and no boundary solutions are observed.

Method	Grid MSE		Stress region MSE		Brier proxy		Log loss	
	Value	Red.	Value	Red.	Value	Red.	Value	Red.
Convex binomial	0.011632	—	0.046939	—	0.011759	—	0.450148	—
Fixed logit-quartic, $\lambda = 1$	0.002098	82.02%	0.008200	82.64%	0.002225	81.13%	0.415273	7.75%
Selected logit-quartic	0.001683	85.49%	0.005914	87.32%	0.001810	84.56%	0.413499	8.14%

Table 7: Binary-response likelihood illustration of the projection-and-pooling framework on the logit scale. “Red.” denotes percentage reduction relative to the convex binomial isotonic benchmark.

F Boundary-avoiding tuning sensitivity in the QQQ illustration

Table 8 reports a sensitivity analysis for the validation-based tuning rule in the main empirical setting ($h = 60$, $G = 8$). We vary the upper bound on admissible non-convexity $\lambda_{\max}$ and the minimum validation-exceedance requirement imposed on the convex 0.99 fit before any positive λ is permitted.

We also vary the minimum required validation improvement threshold (0.25%, 0.5%, and 1.0%). Once the boundary and exceedance filters are fixed, this latter threshold does not materially affect the selected folds. Across all specifications, the selected quartic estimator continues to reduce global grid MSE, stress-region MSE, and stress-region 0.99 pinball loss, while avoiding boundary solutions.

$\lambda_{\max}$	Exceedance floor	Grid MSE	Stress region MSE	Stress region pinball _{0.99}
0.125	0.03	4.45%	6.22%	7.41%
0.125	0.05	3.91%	5.51%	6.35%
0.150	0.03	5.32%	7.49%	8.97%
0.150	0.05	4.64%	6.56%	7.61%
0.200	0.03	5.32%	7.49%	8.97%
0.200	0.05	4.64%	6.56%	7.61%

Table 8: Sensitivity of QQQ results to tuning parameters in the boundary-avoiding validation rule.

G Mechanism-design motivation for the separable objective class

This appendix briefly explains the mechanism-design motivation for the separable objective class as in (1.2) under order constraints. The connection arises through ironing, an order-constrained optimization procedure that produces pooling and, in discrete settings, leads naturally to isotonic regression.

G.1 One-dimensional ironing

Let $\Theta = [0, 1]$ denote a type space, where $\theta \in \Theta$ represents an agent’s private type. Types are distributed according to a distribution function F with density $\tilde{f} > 0$. The decision variable is a monotone allocation rule $\phi(\theta) \in [0, \bar{x}]$, where $\phi(\theta)$ represents an allocation level or allocation probability assigned to type θ . Define the feasible class $\mathcal{M}_1 := \{\phi : \Theta \rightarrow [0, \bar{x}] \mid \phi \text{ is nondecreasing}\}$ for $\bar{x} > 0$. Given a virtual-surplus objective $J : [0, \bar{x}] \times \Theta \rightarrow \mathbb{R}$, the ironing problem studied by [Toikka \(2011\)](#) takes the form

$$\sup_{\phi \in \mathcal{M}_1} \int_0^1 J(\phi(\theta), \theta) dF(\theta).$$

A common specification is

$$J(x, \theta) = k(\theta)a(x) + b(x) + \ell(\theta),$$

where $a(\cdot)$ is strictly increasing. A classical example is the optimal auction formulation of [Myerson \(1981\)](#):

$$k(\theta) = \theta - \frac{1 - F(\theta)}{\tilde{f}(\theta)}, \quad a(x) = x, \quad b(x) = -C(x), \quad \ell(\theta) = 0,$$

where $k(\theta)$ is the virtual valuation and $C(x)$ is the seller’s cost function; see also [Mussa and Rosen](#)

(1978). When $k(\theta)$ fails to be monotone in θ , the monotonicity constraint on ϕ becomes binding and induces pooling across adjacent types. In discrete-type settings, these pooled regions correspond to constant fitted blocks, matching the level-set structure of isotonic regression.

G.2 Multidimensional types and partial orders

When $\theta = (\theta_1, \dots, \theta_d)$ is multidimensional, incentive-compatibility constraints are generally more complex and need not reduce to a single monotonicity condition (Rochet and Stole; 2003). Nevertheless, under additional structure, Bedard et al. (2023) (cf. Rochet and Choné (1998), Bedard et al. (2020)) derive a multidimensional ironing problem in which the allocation rule is required to be nondecreasing with respect to the product partial order

$$\theta^{(1)} \preceq \theta^{(2)} \iff \theta_i^{(1)} \leq \theta_i^{(2)}, \quad i = 1, \dots, d.$$

Let $\Theta = \prod_{i=1}^d [\underline{\theta}_i, \bar{\theta}_i]$ and suppose θ has product density $\tilde{f}(\theta)$. In this setting, the virtual-surplus objective takes the form

$$J(x, \theta) = x \text{MR}(\theta) - C(x),$$

where $\text{MR}(\theta)$ is the multidimensional marginal revenue function. The seller's problem becomes

$$\sup_{\phi \in \mathcal{M}_2} \int_{\Theta} J(\phi(\theta), \theta) \tilde{f}(\theta) d\theta,$$

where $\mathcal{M}_2 = \{\phi : \Theta \rightarrow [0, \bar{x}] \mid \phi \text{ is nondecreasing with respect to } \preceq\}$.

Thus, the feasible set is determined by a partial order on types, while the objective remains separable across indices through the common nonlinear component $C(x)$.

G.3 Discretization and reduction to separable isotonic regression

If each coordinate θ_i takes values on a finite grid, the multidimensional ironing problem reduces to

$$\sup_{\phi \in [0, \bar{x}]^{|\Theta|}} \sum_{\theta \in \Theta} \left[\phi(\theta) \text{MR}(\theta) \tilde{f}(\theta) - C(\phi(\theta)) \tilde{f}(\theta) \right]$$

subject to $\phi(\theta^{(1)}) \leq \phi(\theta^{(2)})$ whenever $\theta^{(1)} \preceq \theta^{(2)}$. Writing $i \equiv \theta$ and defining

$$k_i = \text{MR}(\theta) \tilde{f}(\theta), \quad a(x) = x, \quad m_i = \tilde{f}(\theta), \quad b(x) = -C(x), \quad \ell_i = 0,$$

places the problem directly into the separable form studied in the main text.

This connection motivates the shared nonlinear component structure underlying the projection-and-pooling decomposition. In particular, the common cost component $C(x)$ appears identically across indices, while heterogeneity enters only through the linear coefficients k_i and weights m_i . As a result, multidimensional ironing on a finite grid naturally produces the pooling structure associated with isotonic regression.

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